UTI Microfluidic Cartridge Diagnostic Report

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Patient: Anita Sharma (45y / Female)
Facility: AIIMS New Delhi - Clinical Microbiology Reference Lab
Chief Complaint: Acute pyuria & dysuria evaluation

MRN: MRN-AIIMS-90481 | Specimen: SP-ICMR-2026-041A
Clinician: Dr. Rajesh Malhotra, MD (AIIMS Infectious Diseases)
Collection: Midstream Clean-Catch Urine

DIAGNOSTIC STATUS: ASYMPTOMATIC BACTERIURIA — K. PNEUMONIAE

Pathogen detected (K. pneumoniae) in the absence of significant pyuria (<10 WBC/µL (Normal physiological range)). This pattern is consistent with asymptomatic bacteriuria or epithelial colonization. Routine antimicrobial therapy is generally withheld per stewardship guidelines unless indicated by high-risk conditions (e.g. pregnancy). [HIGH-RISK RECURRENCE PATTERN: Recurrent UTI pattern detected: 12 positive scans recorded in the past 6 months. Repeat infection detected with Klebsiella pneumoniae (5 visits in past 6 months). Resistance change detected: Ciprofloxacin evolved from Sensitive (9/19/2026) to Resistant. Worth discussing antimicrobial prophylaxis or urological evaluation with a clinician.]

1. Pyuria / Pus-Cell Assay (Layer 1 — Leukocyte Esterase)

INFECTION SEVERITY	PYURIA WBC ESTIMATE	SIGNAL OPTICAL DENSITY
NEGATIVE	<10 WBC/µL (Normal physiological range)	0.000 OD

2. 4-Plex Pathogen Aptamer Panel (Layer 2 — Aptamer Capture)

TARGET ORGANISM	GRAM STAIN	OPTICAL DENSITY	STATUS
Escherichia coli	Gram-negative	0.040 (Cutoff: 0.22)	Not Detected
Klebsiella pneumoniae	**Gram-negative**	**0.380 (Cutoff: 0.2)**	**[DETECTED]**
Proteus mirabilis	Gram-negative	0.040 (Cutoff: 0.2)	Not Detected
Pseudomonas aeruginosa	Gram-negative	0.040 (Cutoff: 0.21)	Not Detected

3. Antibiotic Susceptibility Testing (Layer 3 — 4× Replicate Panel)

ANTIMICROBIAL AGENT	CLASS / LINE	MEAN OD ± SD	CONCORDANCE	RESULT
Ampicillin	Aminopenicillin	0.058 ± 0.02	4/4 concordant (100%)	**SENSITIVE**
Ciprofloxacin	Fluoroquinolone	0.780 ± 0.03	4/4 concordant (100%)	**RESISTANT**
Nitrofurantoin* (1st-Line)	Nitrofuran	0.058 ± 0.02	4/4 concordant (100%)	**SENSITIVE**
Cotrimoxazole* (1st-Line)	Folate Synthesis Inhibitor	0.058 ± 0.02	4/4 concordant (100%)	**SENSITIVE**
Fosfomycin* (1st-Line)	Phosphonic Acid Derivative	0.058 ± 0.02	4/4 concordant (100%)	**SENSITIVE**

ANTIMICROBIAL STEWARDSHIP & FIRST-LINE RECOMMENDATIONS:
Sensitive First-Line Options: Nitrofurantoin, Cotrimoxazole. Reserve fluoroquinolones (Ciprofloxacin) to preserve efficacy and prevent resistance escalation.

LEGAL MEDICAL DISCLAIMER: Genex is an investigational point-of-care software reader for microfluidic cartridges. This report does not constitute a certified medical diagnosis and is intended to support clinical decision-making. Treatment must be confirmed by a licensed medical practitioner in accordance with local antimicrobial guidelines.